

VEIL: Variance-Echoed Instance-Less Prompt Learning for Membership-Resistant Federated Vision–Language Models

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Abstract

Federated prompt learning adapts frozen vision–language models through compact updates, yet those updates can preserve instance-specific membership traces. We introduce **VEIL** (Variance-Echoed Instance-Less Prompt Learning), a three-stage client mechanism that converts private embeddings into local class-distribution surrogates before prompt optimization and sanitizes only the resulting prompt delta. Its central operation samples covariance-shaped echoes around class prototypes without forming or communicating a full covariance matrix. We audit six complementary membership attacks at the stringent true-positive rate at 1% false-positive rate (TPR@1%FPR) using label-matched candidates, isolated audit randomness, three datasets, and repeated non-IID partitions. Against FedAvg defenses, **VEIL** attains the lowest cross-dataset mean attack leakage (0.0145) while matching Prompt-DP mean accuracy within 0.12 percentage points. Under the same six attacks, it also exceeds DP-FPL and FedASK accuracy on every dataset and reduces their cross-dataset mean leakage by 3.8% and 36.7%, respectively. DP-FPL remains better on DTD and slightly better in cross-dataset worst-attack leakage, exposing a genuine transfer boundary rather than universal dominance. **VEIL** is therefore an empirical membership defense, not a differential-privacy guarantee.

1 Introduction

Vision–language models such as CLIP provide transferable representations that can be adapted with a small set of learned context tokens rather than full-model fine-tuning (Radford et al. 2021; Zhou et al. 2022). Federated prompt learning combines this parameter efficiency with decentralized optimization: clients retain their images and communicate only prompt updates (McMahan et al. 2017; Zhao et al. 2023; Guo et al. 2024). The resulting protocol is cheap, but it exposes a recurring sequence of low-dimensional, task-specific updates. Such updates and the released prompt can leak whether a candidate example was used for local training, even when the frozen backbone never leaves the server.

Membership inference is particularly consequential at low false-positive rates, where an attacker can make high-confidence claims about a small number of people or records

(Shokri et al. 2017; Carlini et al. 2022). Existing defenses offer incomplete choices. Differentially private optimization gives a formal guarantee, but noise can destabilize prompt learning in strongly non-IID, few-shot regimes (Abadi et al. 2016; Duan et al. 2023). Output-centered methods such as HAMP reduce overconfidence, yet do not directly erase update alignment (Chen and Pattabiraman 2024). Private federated prompt and low-rank training mechanisms, including DP-FPL and FedASK (also called FedSAK in some project documentation), change the communicated parameterization but remain subject to empirical auditing (Tran et al. 2025; Wen et al. 2025).

We ask whether the client can preserve the class structure needed for prompt learning without repeatedly optimizing on identifiable instance directions. Our answer is **VEIL**. Like a veil that preserves a silhouette while hiding identity, the client presents the prompt optimizer with variance-shaped echoes of its class distribution: the semantic outline remains recognizable, but the individual that generated it is obscured. The method operates entirely in the frozen image-embedding space. It compresses local class geometry into an implicit low-rank sampler, learns the prompt from distributional surrogates rather than raw instance anchors, and releases a norm-controlled prompt delta. These three stages target the representation and protocol channels without changing the frozen backbone or server optimizer.

Our contributions are threefold:

- We identify direct instance anchoring as a membership channel in federated soft-prompt tuning and derive an implicit class-geometry sampler that retains local semantic variation without communicating client statistics.
- We develop the three-stage **VEIL** mechanism: private geometry estimation, instance-obscured surrogate learning, and structured prompt release, while keeping CLIP frozen and aggregating only trainable parameters.
- We provide a protocol-aware benchmark that compares **VEIL**, DP-FPL, and FedASK under six attacks with exactly label-matched candidates and reports both per-attack leakage and utility across three non-IID datasets.

2 Related Work

Federated prompt learning. Prompt tuning adapts a pre-trained model through a small trainable context (Lester, Al-

Rfou, and Constant 2021; Zhou et al. 2022). FedAvg supplies the standard weighted aggregation rule for decentralized optimization (McMahan et al. 2017), and FedPrompt and PromptFL demonstrate the communication advantages of exchanging prompts (Zhao et al. 2023; Guo et al. 2024). DP-FPL factorizes global and personalized prompts and perturbs private updates, while FedASK aggregates both low-rank factors through an asymmetric double-sketch mechanism (Tran et al. 2025; Wen et al. 2025). Our work does not replace these mechanisms; it studies a complementary defense at the client representation and upload levels.

Membership inference and auditing. Black-box membership inference exploits differences in model behavior on training and non-training examples (Shokri et al. 2017). Nasr et al. extend the threat to passive and active white-box adversaries (Nasr, Shokri, and Houmansadr 2019). Modern attacks emphasize calibrated likelihood ratios and low-FPR evaluation (Carlini et al. 2022; Zarifzadeh, Liu, and Shokri 2024), while scalable quantile regression learns input-dependent non-member thresholds (Bertran et al. 2023). FedMIA uses other clients as a per-round null population and measures confidence or gradient-update alignment (Zhu et al. 2025). We retain each attack’s information source but specialize features to frozen-backbone prompt tuning.

Membership defenses. DP-SGD clips per-example gradients and adds Gaussian noise (Abadi et al. 2016); private prompt learning applies related ideas to a much smaller parameter space (Duan et al. 2023). HAMP combines high-entropy soft targets, entropy regularization, and low-confidence prediction mapping (Chen and Pattabiraman 2024). In contrast, VEIL intervenes before the prompt loss sees an individual image embedding. Its geometry sampling resembles distributional feature augmentation, but all class statistics remain local and the objective is membership resistance rather than data sharing or domain generalization.

3 Motivation: Compact Updates Still Carry Individual Traces

Freezing CLIP eliminates backbone-gradient leakage but does not eliminate three prompt-specific membership channels. First, the loss and confidence trajectory of a member can remain systematically different over rounds. Second, the gradient induced by a candidate can align with the observed client prompt delta. Third, a final prompt may retain an input-dependent confidence gap. Because the trainable prompt is small, each private mini-batch contributes to the same compact parameter block; parameter efficiency can therefore concentrate rather than remove membership evidence.

A direct response is to perturb the prompt gradient. In few-shot non-IID training, however, individual classes may be represented by only a handful of local examples, so indiscriminate noise can remove useful directions and cause large seed-to-seed utility variation. Our working hypothesis is that prompt learning primarily needs class-level semantic position and local variation, whereas repeatable instance-level directions form an avoidable membership channel. This yields three design requirements: preserve local class geometry

without exporting statistics, remove raw member features from the prompt objective, and control the residual prompt update before aggregation. The mechanism must satisfy all three while keeping the backbone frozen and the standard FedAvg server unchanged.

4 Problem Formulation and Threat Model

There are K clients. Client k owns $D_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ and the server maintains trainable prompt θ^t . The CLIP image encoder ϕ and text encoder ψ are frozen. Given class c , $\psi(c; \theta)$ denotes its normalized text representation. Only tensors whose `requires_grad` flag is true are optimized and aggregated. Under FedAvg, the server computes

$$\theta^{t+1} = \sum_{k \in S_t} \frac{n_k}{\sum_{j \in S_t} n_j} \theta_k^{t+1}. \quad (1)$$

The adversarial server observes protocol messages from all selected clients and released prompt checkpoints, and can query the released model. It does not see raw client images, local feature banks, class means, or covariance factors. For a candidate (x, y) and target client k^* , it predicts whether $(x, y) \in D_{k^*}$. We evaluate at TPR@1%FPR, following low-FPR membership auditing practice (Carlini et al. 2022). VEIL is an empirical defense: its Gaussian upload perturbation is not claimed to satisfy a calibrated (ϵ, δ) -DP bound.

5 VEIL

“Instance-less” refers to the optimization interface, not to data disposal: private examples are used locally to estimate class geometry, but their individual embeddings never enter the prompt loss. Algorithm 1 summarizes the complete mechanism in three stages.

5.1 Stage I: Private Semantic Geometry

Client k first encodes each local image with the frozen image tower and detaches the normalized representation from the computation graph:

$$z_i = \frac{\phi(x_i)}{\|\phi(x_i)\|_2}. \quad (2)$$

For an observed class c , the local semantic center is

$$\mu_{k,c} = \frac{1}{n_{k,c}} \sum_{i: y_i=c} z_i. \quad (3)$$

For $n_{k,c} \geq 2$, rather than forming a dense $d \times d$ covariance, VEIL stores the low-rank centered factor

$$G_{k,c} = \frac{1}{\sqrt{n_{k,c} - 1}} \begin{bmatrix} (z_1 - \mu_{k,c})^\top \\ \vdots \\ (z_{n_{k,c}} - \mu_{k,c})^\top \end{bmatrix}. \quad (4)$$

Its Gram matrix exactly recovers the empirical covariance in feature space:

$$G_{k,c}^\top G_{k,c} = \hat{\Sigma}_{k,c}. \quad (5)$$

The feature bank, class means, and factors remain client-local throughout the round. This stage therefore extracts the class-level signal needed by the prompt while defining an explicit boundary around private instance geometry.

5.2 Stage II: Instance-Obscured Surrogate Learning

For each class label encountered in a mini-batch, the direct image anchor is replaced by its normalized prototype:

$$p_{k,c} = \frac{\mu_{k,c}}{\|\mu_{k,c}\|_2}. \quad (6)$$

To avoid collapsing a heterogeneous class to one point, VEIL draws m geometric echoes. For a fresh standard Gaussian vector $\varepsilon \in \mathbb{R}^{n_{k,c}}$, an echo is

$$\tilde{z}_{k,c} = \frac{\mu_{k,c} + s\varepsilon^\top G_{k,c}}{\|\mu_{k,c} + s\varepsilon^\top G_{k,c}\|_2}. \quad (7)$$

Conditioned on the local bank, the perturbation has covariance proportional to $\hat{\Sigma}_{k,c}$, so it follows the observed class subspace without selecting any member as its center. Classes with fewer than two examples use isotropic noise with standard deviation 0.02. Let $\tilde{B}_k$ contain one prototype and m echoes for every batch label. Only the text prompt remains differentiable, and the client minimizes

$$\mathcal{L}_k(\theta) = \frac{1}{|\tilde{B}_k|} \sum_{(z,y) \in \tilde{B}_k} -\log \frac{\exp(\tau z^\top \psi(y; \theta))}{\sum_{c'} \exp(\tau z^\top \psi(c'; \theta))}. \quad (8)$$

This objective preserves discriminative class geometry while removing the repeatable instance direction exploited by loss- and update-based auditors. We use $m = 3$ and $s = 0.6$ at the selected operating point.

5.3 Stage III: Structured Prompt Release

Surrogate learning suppresses direct anchors but does not prevent a rare local class from shaping the prompt update. Let the trainable local delta be

$$\Delta_k^t = \theta_k^{t+1} - \theta^t. \quad (9)$$

VEIL first applies global norm control:

$$\mathcal{C}_C(\Delta_k^t) = \Delta_k^t \min \left(1, \frac{C}{\|\Delta_k^t\|_2} \right). \quad (10)$$

It then releases the perturbed update

$$\hat{\Delta}_k^t = \mathcal{C}_C(\Delta_k^t) + \xi_k^t. \quad (11)$$

The release noise follows

$$\xi_k^t \sim \mathcal{N}(0, (\rho C)^2 I). \quad (12)$$

The unchanged server aggregates only these detached trainable deltas:

$$\theta^{t+1} = \theta^t + \sum_{k \in S_t} \frac{n_k}{\sum_{j \in S_t} n_j} \hat{\Delta}_k^t. \quad (13)$$

We use $C = 0.5$ and $\rho = 0.11$. Randomness is keyed by client, round, and purpose, and every released state is a detached tensor copy. The evaluation protocol additionally applies a fixed temperature $T = 4$ to final-query logits; because positive scaling preserves score order, we treat this as

Algorithm 1 Three-Stage VEIL Client Update

- Require:** Frozen image encoder ϕ , prompt θ^t , local data D_k
- 1: **Geometry:** build detached $(\mu_{k,c}, G_{k,c})$ using Eqs. (3)–(4)
 - 2: **Surrogates:** optimize only θ on prototypes and echoes using Eqs. (7)–(8)
 - 3: **Release:** upload detached $\theta^t + \hat{\Delta}_k^t$ using Eqs. (10)–(11)
-

release calibration rather than a distinct methodological contribution. The ablation confirms that it does not change the reported rank-based metrics.

The additional feature-bank pass requires one frozen image-encoder evaluation per local example per round. For embedding dimension d , the detached bank costs $O(n_k d)$ memory and echo generation costs $O(m n_b d)$ per batch of size n_b . No class statistic, extra server state, or additional communication is required. The mechanism therefore changes what supervises the prompt and how its delta is released, while preserving the original federated interface.

6 Experimental Methodology

6.1 Research Questions

We ask three questions. **RQ1:** Does VEIL improve the privacy–utility trade-off over defenses attached to the same FedAvg backbone? **RQ2:** How does VEIL compare with DP-FPL and FedASK/FedSAK under each of six membership attacks? **RQ3:** Which of the three stages controls the observed trade-off?

6.2 Datasets and Federated Protocol

We use Flowers102 (Nilsback and Zisserman 2008), Caltech101 (Fei-Fei, Fergus, and Perona 2004), and DTD (Cimpoi et al. 2014). They respectively test fine-grained appearance, heterogeneous objects, and material texture. Every dataset uses 16 labeled examples per class from its training partition, while evaluation uses the complete held-out test partition; DTD uses its first predefined fold. We partition the few-shot training set over 10 clients with a Dirichlet distribution of concentration $\alpha = 0.1$, select all 10 clients per round, and train for five global rounds with one local epoch. The batch size is 16, learning rate is 10^{-3} , and prompt length is 16. Standard prompt updates use SGD with zero momentum and weight decay; private mechanisms use their method-specific update rules at the same learning rate. The audit target is client 0. Reported aggregates use seeds 42–44 unless stated otherwise. Formal runs use two NVIDIA RTX 4090 GPUs; CLIP ViT-B/32 is loaded from local storage and remains frozen. The host uses an Intel Core i5-14600KF, two 24-GB RTX 4090 GPUs, Linux 6.8, Python 3.9.21, PyTorch 2.7.1 with CUDA 12.8, Transformers 4.57.3, and NumPy 1.26.3. The main matrix contains 54 independent runs (three datasets, three seeds, and six mechanisms); the component study adds seven Flowers102 seed-42 runs. Each process occupies one GPU, and no process silently falls back to CPU.

We tune VEIL on Flowers102 seeds 42–43 and reserve seed 44 for replication. The selection rule minimizes worst-

Dataset	Classes	Train	Test	Mean/client
Flowers102	102	1,632	6,149	163.2
Caltech101	101	1,616	5,647	161.6
DTD	47	752	1,880	75.2

Table 1: Dataset scale after the common 16-shot-per-class protocol. The last column is the arithmetic mean over 10 clients; actual client counts are unequal under the $\alpha = 0.1$ Dirichlet partition. Test counts cover the complete evaluation partitions.

attack TPR@1%FPR subject to mean development accuracy of at least 0.60, with accuracy breaking ties. We then freeze all hyperparameters for Caltech101 and DTD; the complete sweep is provided in the artifact rather than repeated in the main text. All split, initialization, client, and audit generators are keyed by the reported seed. DP mechanisms keep `reproducible_dp_noise=false`; their Gaussian draws intentionally use a private non-deterministic generator, so seed labels identify the repeated scenario rather than fixing privacy noise byte-for-byte.

6.3 Baselines

The shared-backbone study compares no defense, Prompt-DP, HAMP, and VEIL. The private-mechanism study compares VEIL with DP-FPL and FedASK (named FedSAK in some project documentation). Method-specific clipping, noise, local steps, and factor ranks follow the aligned public-repository configurations; data, optimization budget, candidate construction, and audit fields remain fixed. Prompt-DP is applied only to trainable CLIP prompt tensors. Because a non-differentiable randomized HAMP release would make gradient-based auditors undefined, we use its high-entropy training objective and a deterministic, argmax-preserving temperature map; this is reported as a prompt-learning adaptation rather than a byte-identical reproduction. Likewise, DP-FPL’s global/local factors and FedASK’s two-stage sketches are mapped to CLIP prompt matrices; their original language-model tasks are not reproduced. FedAvg, Prompt-DP, HAMP, VEIL, and FedASK release one shared prompt. DP-FPL retains its defining persistent client-local residual and is evaluated with each client’s own global-plus-local prompt; accuracies are then micro-averaged over the same client test partitions. Forcing DP-FPL to discard its local state would no longer reproduce its stated algorithm.

6.4 Attacks and Metrics

We evaluate FedMIA confidence, FedMIA gradient-update cosine, their fixed joint score, Nasr-style passive white-box inference, offline RMIA, and quantile- regression MIA. Together they cover output confidence, update alignment, white-box prompt statistics, reference-model likelihood ratios, and learned input-dependent thresholds. Exact attack parameters are fixed before the main matrix and recorded in the artifact. Each audit begins with 64 members and 64 non-members. Their label histograms are exactly equal. Non-members are drawn first from held-out client test partitions

and, if a class is under-supplied, from other clients’ training partitions; both are non-members with respect to the target client. A run fails if this combined pool cannot meet the constraint. A separate generator is supplied to every audit DataLoader, so auditing cannot advance the training RNG.

Our primary privacy metric is TPR@1%FPR (lower is better). We additionally report the maximum TPR@1%FPR over attacks for each run, the unweighted mean over attacks, and AUC as a diagnostic. Clean utility is final test accuracy. Tables report mean and sample standard deviation across seeds. Since three seeds and finite candidate pools provide limited power, we avoid significance claims and expose per-seed variation. With at most 64 evaluated non-members, a single false positive already exceeds 1% FPR; TPR@1%FPR therefore uses the empirical zero- false-positive threshold and is visibly quantized. We retain the conventional metric for comparability but interpret exact zeros cautiously.

7 Results

7.1 FedAvg Defense Comparison

Table 2 answers RQ1 under a shared training backbone. We use a single table because it exposes the exact utility and both leakage summaries more directly than a redundant privacy-utility scatter plot.

VEIL yields the lowest mean leakage on Flowers102 and both best privacy summaries on Caltech101, while retaining substantially more utility than HAMP on Flowers102. DTD is the counterexample: Prompt-DP is better on both privacy summaries, and VEIL loses 9.10 accuracy points relative to no defense. Averaged equally over datasets, VEIL reaches 0.6001 accuracy and 0.0145 mean TPR@1%FPR, 15.3% and 24.2% below Prompt-DP and HAMP in mean leakage. The dataset-level reversal is why we avoid a universal superiority claim.

7.2 Comparison with DP-FPL and FedASK under Six Attacks

RQ2 compares all three private training mechanisms under the same attacks. Figure 1 retains the full attack vector, while Table 3 summarizes the associated privacy-utility trade-off.

On Flowers102, VEIL is best or tied on five of six attacks and improves all three table metrics. On Caltech101, it is best on FedMIA-Cosine, FedMIA-Joint, and RMIA; it matches DP-FPL’s mean leakage while adding 6.12 accuracy points. Across datasets, VEIL averages 0.6001 accuracy, compared with 0.5512 for DP-FPL and 0.4888 for FedASK. Its mean attack leakage is 3.8% below DP-FPL and 36.7% below FedASK. These are the principal advantages over the two private baselines.

The full profile also reveals the boundary. On DTD, DP-FPL has lower mean and worst leakage; VEIL wins only FedMIA-Joint and is particularly exposed to RMIA. Accordingly, DP-FPL’s cross-dataset worst leakage (0.0451) is slightly below VEIL (0.0469), even though VEIL has the stronger utility and mean-leakage averages. This attack-dependent result would disappear in a single scalar ranking.

Dataset	Defense	Utility $\uparrow$	Membership leakage $\downarrow$	
		Accuracy	Worst TPR@1%FPR	Mean TPR@1%FPR
Flowers102	No defense	0.6609 $\pm$ 0.0030	0.0469 $\pm$ 0.0271	0.0191 $\pm$ 0.0066
	Prompt-DP	0.6170 $\pm$ 0.0207	0.0573 $\pm$ 0.0090	0.0191 $\pm$ 0.0080
	HAMP	0.5471 $\pm$ 0.0120	0.0313 $\pm$ 0.0271	0.0122 $\pm$ 0.0143
	VEIL	0.6201 $\pm$ 0.0013	0.0573 $\pm$ 0.0239	0.0113 $\pm$ 0.0066
Caltech101	No defense	0.8677 $\pm$ 0.0037	0.0573 $\pm$ 0.0738	0.0148 $\pm$ 0.0157
	Prompt-DP	0.8112 $\pm$ 0.0382	0.0781 $\pm$ 0.0716	0.0278 $\pm$ 0.0330
	HAMP	0.7971 $\pm$ 0.0115	0.1042 $\pm$ 0.1273	0.0321 $\pm$ 0.0296
	VEIL	0.8040 $\pm$ 0.0070	0.0417 $\pm$ 0.0451	0.0130 $\pm$ 0.0158
DTD	No defense	0.4672 $\pm$ 0.0093	0.0365 $\pm$ 0.0090	0.0130 $\pm$ 0.0045
	Prompt-DP	0.3755 $\pm$ 0.0289	0.0156 $\pm$ 0.0000	0.0043 $\pm$ 0.0015
	HAMP	0.4323 $\pm$ 0.0027	0.0312 $\pm$ 0.0000	0.0130 $\pm$ 0.0026
	VEIL	0.3762 $\pm$ 0.0231	0.0417 $\pm$ 0.0090	0.0191 $\pm$ 0.0030

Table 2: FedAvg defenses under identical data, optimization, participation, and audit settings. Values are mean $\pm$ sample standard deviation over seeds 42–44. Bold marks the best value within each dataset; shaded rows denote our method.

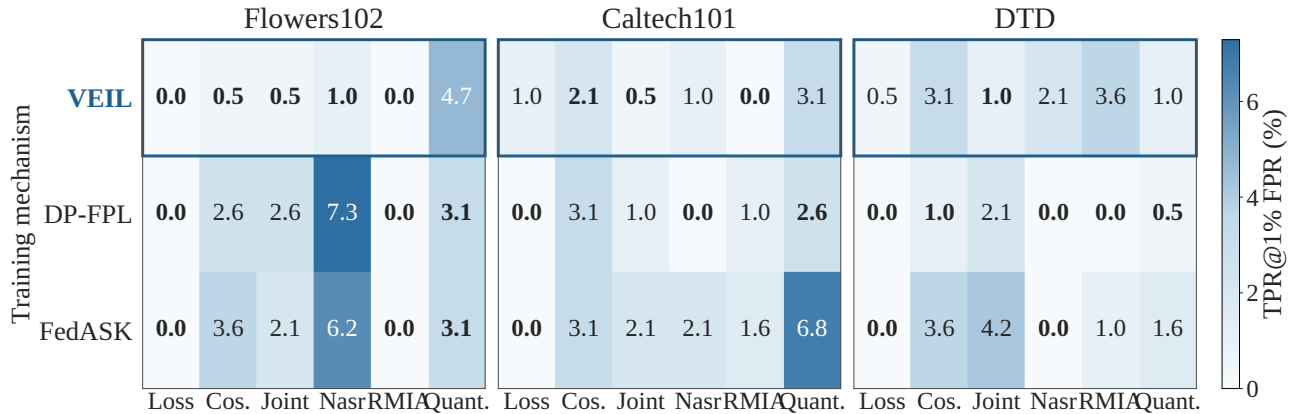


Figure 1: Attack-specific comparison of VEIL, DP-FPL, and FedASK, averaged over seeds 42–44. All panels share one zero-anchored color scale; lower is better, and cell values are percentages. Bold cells mark the lowest value per attack (including ties), and the outlined row denotes our method.

7.3 Component Analysis

The focused RQ3 ablation is clearer as exact values than as three stacked bar plots. Table 5 varies one component at a time on Flowers102 seed 42.

Upload smoothing is the clearest trade-off lever: removing it adds 2.99 accuracy points but doubles worst leakage and more than triples mean leakage. Returning to individual anchors doubles mean leakage, while removing geometric echoes raises it by 50%. Removing output calibration changes no rank-based metric, supporting its treatment as protocol calibration rather than a fourth method stage. Because this is a single-seed diagnostic, it identifies mechanisms but does not support a significance claim.

8 Limitations

VEIL provides empirical resistance to the audited attacks, not formal differential privacy. A low measured TPR@1%FPR can reflect finite candidate resolution—including the zero-

false-positive threshold described above—and no finite attack suite proves the absence of leakage. Our experiments freeze one CLIP architecture, use full client participation, and consider three image datasets; partial participation, larger backbones, other modalities, and adaptive attackers remain open. Prototype replacement can also under-represent rare within-class modes, while Gaussian upload smoothing may be inappropriate when downstream users require deterministic updates. The DTD result is a concrete transfer failure: Flowers102-tuned VEIL reduces accuracy without improving either privacy summary, so cross-dataset retuning or adaptive scaling remains necessary.

9 Ethical Statement

The intended benefit is reducing disclosure of whether a person’s example was used during collaborative adaptation. The same mechanism could conceal data provenance or complicate accountability if deployed without audit logs. We there-

Dataset	Method	Utility $\uparrow$	Membership leakage $\downarrow$	
		Accuracy	Worst TPR@1%FPR	Mean TPR@1%FPR
Flowers102	DP-FPL	0.5552 $\pm$ 0.1055	0.0781 $\pm$ 0.0156	0.0260 $\pm$ 0.0094
	FedASK	0.5270 $\pm$ 0.1426	0.0625 $\pm$ 0.0312	0.0252 $\pm$ 0.0128
	VEIL	0.6201 $\pm$ 0.0013	0.0573 $\pm$ 0.0239	0.0113 $\pm$ 0.0066
Caltech101	DP-FPL	0.7428 $\pm$ 0.0486	0.0365 $\pm$ 0.0090	0.0130 $\pm$ 0.0052
	FedASK	0.6063 $\pm$ 0.2308	0.0781 $\pm$ 0.0413	0.0260 $\pm$ 0.0156
	VEIL	0.8040 $\pm$ 0.0070	0.0417 $\pm$ 0.0451	0.0130 $\pm$ 0.0158
DTD	DP-FPL	0.3557 $\pm$ 0.0272	0.0208 $\pm$ 0.0239	0.0061 $\pm$ 0.0066
	FedASK	0.3330 $\pm$ 0.0268	0.0521 $\pm$ 0.0549	0.0174 $\pm$ 0.0183
	VEIL	0.3762 $\pm$ 0.0231	0.0417 $\pm$ 0.0090	0.0191 $\pm$ 0.0030

Table 3: Utility and aggregate leakage for the three mechanisms in Figure 1. Values are mean $\pm$ sample standard deviation over seeds 42–44. Bold marks the best value within each dataset; shaded rows denote our method.

Method	Scope	ϵ upper bound	δ
Prompt-DP	prompt update	46.51–81.51	10^{-5}
DP-FPL	local	122.62	10^{-5}
DP-FPL	global	511.51	10^{-5}
FedASK	local	61.51	10^{-5}

Table 4: Conservative Gaussian-RDP accounting over all nine runs per mechanism. No client or example subsampling amplification is claimed. Prompt-DP varies with the maximum client step count; the other reported bounds are constant. These large upper bounds do not establish strong privacy at the selected noise settings.

Variant	Acc. $\uparrow$	Worst $\downarrow$	Mean $\downarrow$
Full VEIL	0.6216	0.0313	0.0052
Individual anchor	0.6209	0.0313	0.0104
No echoes	0.6209	0.0313	0.0078
No prototype branch	0.6212	0.0313	0.0052
No upload smoothing	0.6515	0.0625	0.0182
No output calib.	0.6216	0.0313	0.0052

Table 5: Component analysis on Flowers102 seed 42. Privacy columns report TPR@1%FPR. Bold marks the best value including ties; the shaded row is the full method.

fore recommend reporting both empirical attack results and any formal privacy accounting, preserving model/data documentation, and avoiding claims that federated learning alone guarantees privacy.

10 Conclusion

We presented VEIL, a three-stage defense for federated CLIP prompt learning: estimate private semantic geometry, optimize on instance-obscured surrogates, and release a norm-controlled prompt delta. This interface removes individual embeddings from the prompt loss without communicating class statistics or changing the FedAvg server. Under six label-matched attacks, VEIL provides the strongest cross-

dataset accuracy and mean-leakage averages among VEIL, DP-FPL, and FedASK, while DP-FPL remains preferable on DTD privacy and slightly better in worst-attack leakage overall. The result supports distributional surrogate learning as a practical empirical complement to formally accounted mechanisms, not as a substitute for differential privacy.

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1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) **yes**
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) **yes**
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) **yes**

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) **no**

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) **yes**

- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) **no**
- 2.4. Proofs of all novel claims are included (yes/partial/no) **no**
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) **yes**
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) **yes**
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) **NA**
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) **NA**

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) **yes**

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) **yes**
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) **NA**
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **NA**
- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) **yes**
- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) **yes**
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) **NA**

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) **yes**

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) **yes**
- 4.3. Any code required for pre-processing data is included

in the appendix (yes/partial/no) **partial**

- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) **partial**
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) **yes**
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) **partial**
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) **yes**
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) **yes**
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) **yes**
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) **yes**
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) **yes**
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) **no**
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) **yes**